

# OUALID OBBAD

Software Engineer | Backend Development

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## PROFESSIONAL SUMMARY

Software engineering student at 1337 / 42 Network (UM6P), focused on backend development with C, C++ and Java. Builds real systems from scratch, including web servers, a Unix shell and containerized applications, with a solid foundation in networking, databases and Linux. Seeking a backend engineering internship.

## EDUCATION

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|------------------------------------------------------------------------|----------------|
| 1337 Coding School (42 Network) – UM6P, Rabat                          | 2024 – Present |
| BTS – Advanced Technician Certificate in Computer Systems and Networks | 2024           |
| Baccalaureate – Electrical Science and Technology                      | 2022           |

## TECHNICAL SKILLS

**Languages:** C, C++ (C++98), Java, Bash  
**Backend:** Spring, Spring JDBC, JDBC, Maven, HikariCP, HTTP/1.1, REST, CGI  
**Databases:** SQL, MySQL, PostgreSQL, MariaDB, HSQLDB, schema design, normalization  
**Systems & Networking:** Linux, POSIX threads, processes, signals, TCP/IP sockets, epoll, IP addressing, subnetting (CIDR), routing  
**DevOps & Tools:** Docker, Docker Compose, NGINX, TLS/SSL, Git, Make, Valgrind  
**Testing:** JUnit 5, Mockito, integration testing

## PROJECTS

### Webserve – HTTP/1.1 Web Server | C++98, Sockets, epoll, CGI – team of 2

- NGINX-inspired HTTP server written from scratch, built on a single-threaded non-blocking epoll event loop with GET, POST and DELETE, virtual hosting, CGI execution, file uploads and a custom configuration-file parser.

### Java Backend Suite | Java, PostgreSQL, JDBC, HikariCP, Spring, JUnit 5, Mockito

- Layered chat application over a normalized PostgreSQL schema, with a JDBC repository layer, connection pooling and Maven packaging.
- Custom mini-ORM built with Java reflection and annotations, backed by a JUnit 5 and Mockito test suite and Spring dependency injection.

### Minishell – Unix Shell | C, Processes, Signals – team of 2

- Bash-like shell built without parsing libraries, structured as a lexer, parser and executor pipeline with pipes, redirections, heredocs, variable expansion, seven built-ins and Bash-accurate signal handling.

### Inception – Containerized Infrastructure | Docker, Docker Compose, NGINX, MariaDB, Redis, TLS

- 8-service Docker stack with hand-written Dockerfiles, fronted by an NGINX TLS reverse proxy as the sole entry point, with network isolation, Docker secrets and persistent volumes.

### cub3D – 3D Raycasting Engine | C, MiniLibX

- Real-time textured 3D renderer built on the DDA raycasting algorithm, with a validating scene-file parser and per-direction texture mapping.

### Philosophers – Concurrency | C, pthreads, Mutexes

- Multithreaded simulation solving deadlock and starvation with mutex-protected shared state and a dedicated monitoring thread.

### ft\_transcendence – Full-Stack Real-Time Platform (In Progress) | Docker, WebSockets, REST, PostgreSQL, HTTPS – team of 4

- Developing a containerized multi-user web application orchestrated with Docker for single-command deployment, serving HTTPS REST APIs and real-time WebSocket pipelines.
- Building backend services with secure user authentication, end-to-end input validation, and a relational database schema optimized for multi-client concurrency.

## LANGUAGES

English: Intermediate    French: Intermediate